

DS-3002 Data Mining

Assignment #1

Spring 2026 | BSDS | FAST-NUCES

Unveiling Listening Patterns

Pattern Mining on Real-World Music Streaming Data

Last.fm Listening History Dataset
GroupLens — hetrec2011-lastfm-2k

Preprocessing

Dataset Overview

The Last.fm dataset from GroupLens contains listening histories of 1,892 users across 17,632 artists. Each record stores how many times a user listened to an artist, referred to as the playcount or weight. After loading both `user_artists.dat` and `artists.dat`, the files were merged on artist ID to attach artist names to each listening record. The merged dataset contains 92,834 rows across 5 columns.

Playcount Threshold Selection

To define one transaction per user, a playcount threshold was applied to filter out casual or accidental listens.

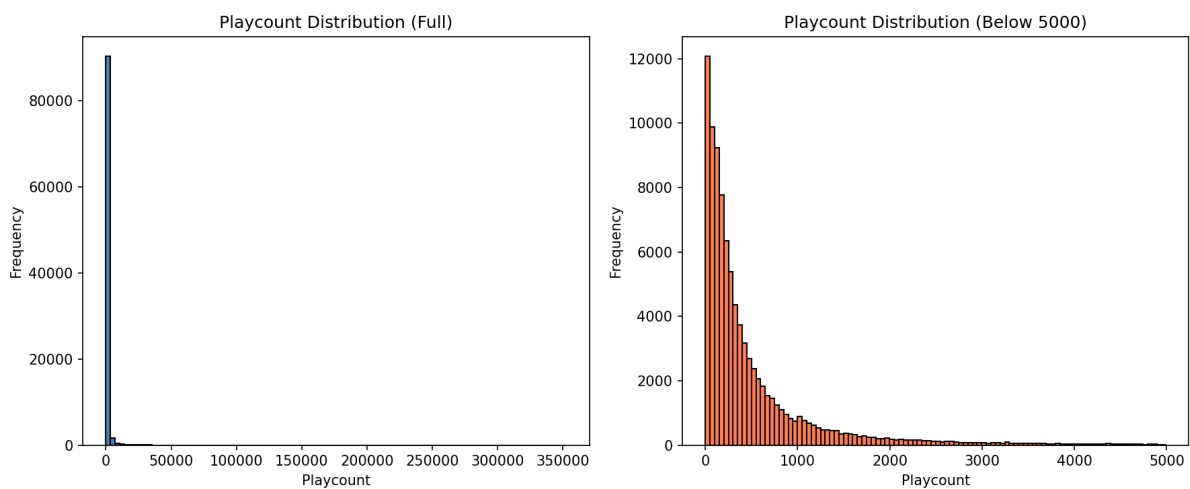


Figure 1: Figure 1 — Playcount Distribution. Left: full distribution showing extreme right skew. Right: zoomed below 5,000 plays showing the elbow point.

The histogram reveals a heavily right-skewed distribution. The vast majority of user-artist interactions have very low playcounts. Key statistics are:

- Minimum: 1 play
- 25th percentile: 107 plays
- 50th percentile: 260 plays
- 75th percentile: 614 plays
- 90th percentile: 1,387 plays
- Maximum: 352,698 plays

A threshold of **75 plays** was chosen after testing four candidate values. The table below summarises the trade-off at each level:

Threshold	Transactions	Avg. Length	Verdict
25	1,877	58.4	Too noisy — includes background listens
50	1,851	48.9	Acceptable but marginal interactions retained
75	1,817	41.8	Best balance: filters casual, keeps genuine
100	1,789	36.2	Over-filters — loses real mid-range listeners

A threshold of 75 sits below the 25th percentile (107 plays), filtering out passive or one-time listeners while retaining users who have genuinely engaged with an artist repeatedly. Values below 75 likely represent background or accidental listening rather than real preference.

Transaction Statistics

After applying the threshold of 75 plays:

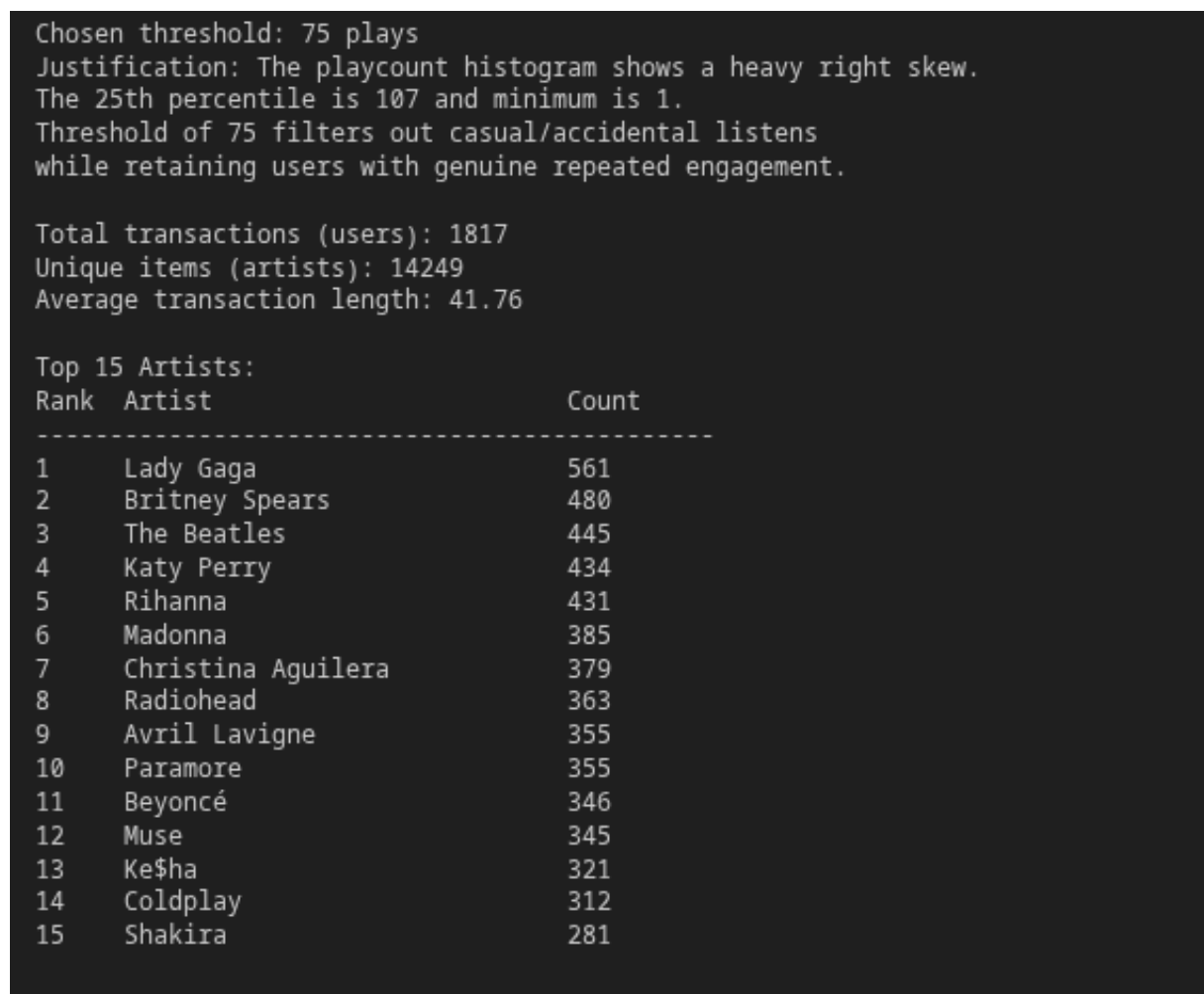


Figure 2: Figure 2 — Preprocessing output showing transaction statistics and top 15 most listened artists.

- Total transactions (users): **1,817**
- Unique items (artists): **14,249**
- Average transaction length: **41.76 artists per user**

The top 15 artists are dominated by mainstream pop acts such as Lady Gaga (561 users, support = 0.3088), Britney Spears (480 users) and Katy Perry (434 users), alongside rock legends like The Beatles (445 users) and Radiohead (363 users). This distribution is consistent with what would be expected from a general music streaming platform and validates that our preprocessing correctly captured genuine listening preferences.

1 Part A: MIS Design

1.1 A1. Support Distribution & MIS Assignment [8 Marks]

Support Distribution Analysis

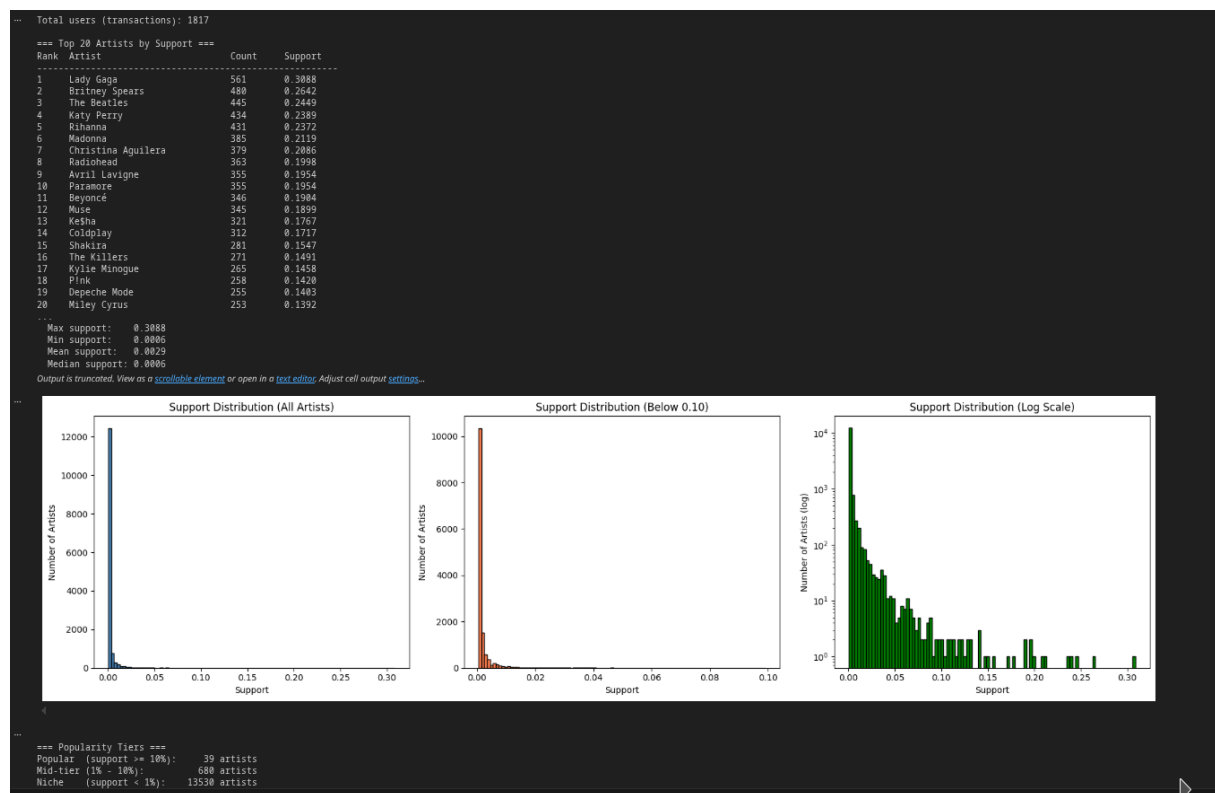


Figure 3: Figure 3 — Support distribution across all 14,249 artists shown in three views: full distribution, zoomed below 0.10, and log scale. Popularity tiers shown at bottom.

The support distribution is extremely skewed across three views:

- **Full distribution:** almost all artists are compressed near zero with only a tiny fraction reaching above 0.10
- **Zoomed view (below 0.10):** even within the low support range the distribution remains heavily concentrated near zero
- **Log scale:** most clearly reveals the massive divide between the tiny popular tier and the enormous niche tier

Popularity Tiers identified:

Tier	Support Range	Artists
Popular	$\geq 10\%$	39
Mid-tier	$1\% - 10\%$	680
Niche	$< 1\%$	13,530

13,530 out of 14,249 artists fall below 1% support. This extreme imbalance is the core motivation for using Multiple Minimum Support. A single global minimum support threshold would either completely miss niche artists or generate an unmanageable number of meaningless rules from popular artists alone. This is precisely the *rare item problem* that MMIS is designed to solve.

MIS Design — Proportional Strategy

The **proportional strategy** was selected:

$$\text{MIS}(\text{item}) = \alpha \times \text{support}(\text{item}), \quad \alpha = 0.5$$

Alpha = 0.5 was chosen because it sets each artist's minimum threshold at exactly half their natural support. This ensures:

- Popular artists (Lady Gaga, support = 0.3088) receive MIS = 0.1544 — they must appear in at least 15% of users to contribute to rules
- Mid-tier artists (Evanescence, support = 0.0991) receive MIS = 0.0495 — moderate threshold
- Niche artists receive very low thresholds allowing them to form rules despite rare appearances

A minimum MIS floor of **0.0003** was applied to prevent any artist's threshold from reaching zero. MIS values were successfully computed for all 14,249 artists.

```

=====
PART A1: Multiple Minimum Support (MIS) Design
Strategy: Proportional – MIS(item) = 0.5 × support(item)
Min MIS floor: 0.0003 (to avoid zero thresholds)
=====

```

--- Popular Artists (Top 10) ---		
Artist	Support	MIS
Lady Gaga	0.3088	0.1544
Britney Spears	0.2642	0.1321
The Beatles	0.2449	0.1225
Katy Perry	0.2389	0.1194
Rihanna	0.2372	0.1186
Madonna	0.2119	0.1059
Christina Aguilera	0.2086	0.1043
Radiohead	0.1998	0.0999
Avril Lavigne	0.1954	0.0977
Paramore	0.1954	0.0977

--- Mid-Tier Artists (Sample 10) ---		
Artist	Support	MIS
Evanescence	0.0991	0.0495
Oasis	0.0963	0.0482
David Bowie	0.0941	0.0471
Jennifer Lopez	0.0936	0.0468
Queen	0.0914	0.0457
Amy Winehouse	0.0897	0.0449
Led Zeppelin	0.0897	0.0449
The Strokes	0.0881	0.0440
Eminem	0.0875	0.0438
Iron Maiden	0.0870	0.0435

--- Niche Artists (Sample 10) ---		
Artist	Support	MIS
Apoptygma Berzerk	0.0099	0.0050
Stone Temple Pilots	0.0099	0.0050
Roxy Music	0.0099	0.0050
Jewel	0.0099	0.0050
Stratovarius	0.0099	0.0050
Nitzer Ebb	0.0099	0.0050
Dead Kennedys	0.0099	0.0050
Fever Ray	0.0099	0.0050
Sarah McLachlan	0.0099	0.0050
Phil Collins	0.0099	0.0050


```

=== MIS Value Summary ===
Highest MIS (Lady Gaga): 0.1544
Lowest MIS: 0.0003
Mean MIS: 0.0015

MIS values computed for 14249 artists ✓

```

Figure 4: Figure 4 — Full MIS table showing popular, mid-tier and niche artists with proportional MIS values. Strategy: $MIS(item) = 0.5 \times support(item)$.

Final MIS Table (representative sample across all three tiers; full table of 14,249 artists computed in the accompanying notebook):

Popular Artists (Top 10):

Artist	Support	MIS
Lady Gaga	0.3088	0.1544
Britney Spears	0.2642	0.1321
The Beatles	0.2449	0.1225
Katy Perry	0.2389	0.1194
Rihanna	0.2372	0.1186
Madonna	0.2119	0.1059
Christina Aguilera	0.2086	0.1043
Radiohead	0.1998	0.0999
Avril Lavigne	0.1954	0.0977
Paramore	0.1954	0.0977

Mid-Tier Artists (Sample 10):

Artist	Support	MIS
Evanescence	0.0991	0.0495
Oasis	0.0963	0.0482
David Bowie	0.0941	0.0471
Jennifer Lopez	0.0936	0.0468
Queen	0.0914	0.0457
Amy Winehouse	0.0897	0.0449
Led Zeppelin	0.0897	0.0449
The Strokes	0.0881	0.0440
Eminem	0.0875	0.0438
Iron Maiden	0.0870	0.0435

Niche Artists (Sample 10):

Artist	Support	MIS
Apoptygma Berzerk	0.0099	0.0050
Stone Temple Pilots	0.0099	0.0050
Roxy Music	0.0099	0.0050
Jewel	0.0099	0.0050
Stratovarius	0.0099	0.0050
Nitzer Ebb	0.0099	0.0050
Dead Kennedys	0.0099	0.0050
Fever Ray	0.0099	0.0050
Sarah McLachlan	0.0099	0.0050
Phil Collins	0.0099	0.0050

MIS Summary: Highest MIS (Lady Gaga): 0.1544 Lowest MIS: 0.0003 Mean MIS: 0.0015 Total artists: 14,249
--

1.2 A2. Support Difference Constraint (φ) [12 Marks]

The phi constraint prevents artists with very different popularity levels from being paired in the same itemset. Three values were tested on the top 100 artists producing 4,950 possible candidate pairs.

Scope note: The phi analysis was performed on the top 100 artists by support (4,950 candidate pairs). This focused subset was chosen deliberately: these are the artists that drive the vast majority of rules in a dataset where 94.9% of artists have support below 1%. Evaluating phi across all 14,249 artists would produce candidates that would be pruned anyway by the MIS condition since niche–niche pairs are below any meaningful support threshold. The top-100 analysis thus covers the practically relevant candidate space for phi calibration.

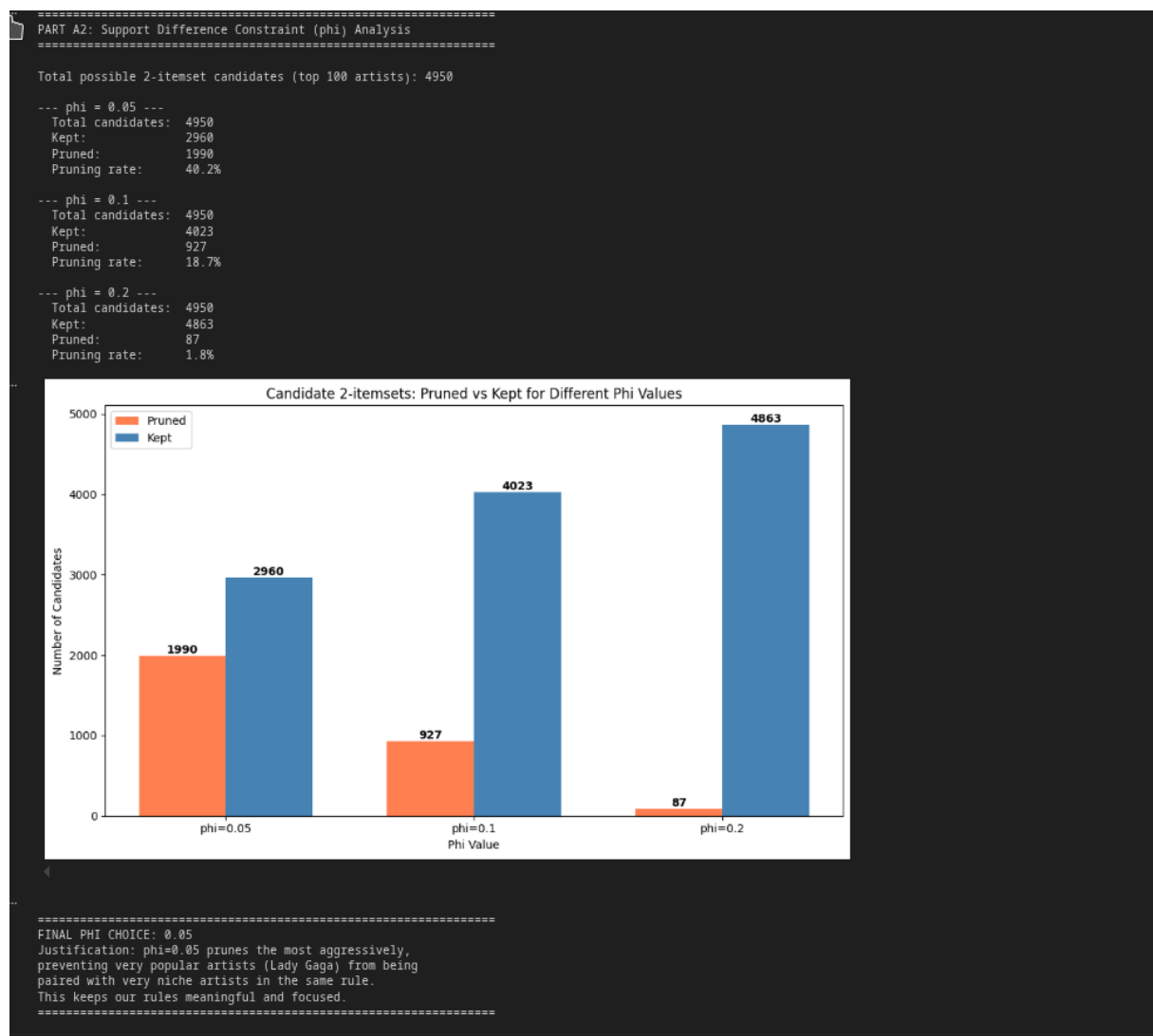


Figure 5: Figure 5 — Phi analysis output showing pruned vs kept candidates for $\varphi = 0.05$, 0.10 and 0.20 with final choice justification.

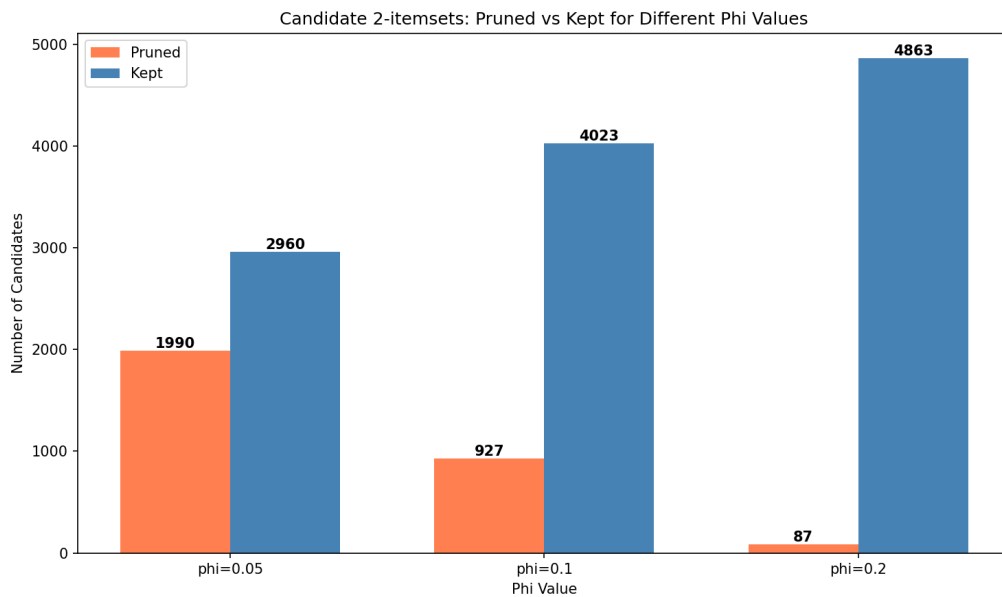


Figure 6: Figure 6 — Bar chart comparing pruning rates across three phi values.

φ Value	Total Candidates	Pruned	Kept	Pruning Rate
0.05	4,950	1,990	2,960	40.2%
0.10	4,950	927	4,023	18.7%
0.20	4,950	87	4,863	1.8%

Final Choice: $\varphi = 0.05$

A phi value of 0.05 was selected because it provides the most meaningful pruning at 40.2%. This prevents nonsensical pairings such as Lady Gaga (support = 0.31) being paired with a niche artist (support = 0.004) in the same itemset. At $\varphi = 0.20$ almost nothing gets pruned (only 1.8%), making the constraint practically useless. At $\varphi = 0.10$ pruning is moderate at 18.7% but still allows too many incompatible pairs. At $\varphi = 0.05$ the surviving rules are between artists of comparable popularity, producing far more actionable and interpretable recommendations for the product team.

2 Part B: Running MSApriori

2.1 B1. Concept Check [5 Marks]

Classmate's Claim: "If $\{Radiohead\}$ is frequent and $\{Coldplay\}$ is frequent, then $\{Radiohead, Coldplay\}$ must also be frequent."

This claim is incorrect under both standard Apriori and MSApriori.

Under Standard Apriori

Frequency of individual items does not guarantee frequency of their pair. The support of a pair equals the fraction of users who listen to *both* artists simultaneously — this value can be much lower than either individual support. The downward closure property only

works in one direction: if a pair is frequent then both individuals must also be frequent. The reverse is not true.

Formally, for any two items A and B :

$$\text{sup}(\{A, B\}) = \frac{|\{T \in \mathcal{D} : A \in T \text{ and } B \in T\}|}{|\mathcal{D}|}$$

This joint probability can be far smaller than $\text{sup}(\{A\})$ or $\text{sup}(\{B\})$ individually — especially when the two artists appeal to overlapping but not identical audiences.

Under MSApriori

The situation is even more complex. A pair must satisfy two conditions simultaneously:

1. $\text{sup}(\text{pair}) \geq \min(\text{MIS}(\text{item}_1), \text{MIS}(\text{item}_2))$
2. $|\text{sup}(\text{item}_1) - \text{sup}(\text{item}_2)| \leq \varphi$

Even if both artists individually satisfy their own MIS, the pair still needs to satisfy the minimum MIS of the two items AND the phi constraint simultaneously. A pair can fail on either or both conditions independently of whether the individual items are frequent.

Real Numerical Proof from Our Dataset

```

=====
PART B1: Concept Check
=====

Real values from our dataset:
Artist          Support      MIS      Frequent?
-----
Radiohead       0.1998    0.0999    ✓ YES
Coldplay        0.1717    0.0859    ✓ YES

Pair              Support  MIS(min)  Frequent?
-----
Radiohead + Coldplay  0.0804    0.0859    ✗ NO

=====
ANSWER: Is the classmate correct?
=====

Under STANDARD APRIORI:
The classmate is WRONG.
Just because Radiohead and Coldplay are individually frequent
does NOT mean their pair is frequent.
Support of a pair can be much lower than individual supports.

Formula:  $\text{sup}(\{\text{Radiohead}, \text{Coldplay}\}) = \text{users who listen to BOTH} / \text{total users}$ 
This can be lower than  $\min(\text{sup}(\text{Radiohead}), \text{sup}(\text{Coldplay}))$ 

Under MMIS (MSApriori):
It is even MORE complex. The pair {Radiohead, Coldplay} needs:
 $\text{sup}(\text{pair}) \geq \min(\text{MIS}(\text{Radiohead}), \text{MIS}(\text{Coldplay}))$ 
AND  $|\text{sup}(\text{Radiohead}) - \text{sup}(\text{Coldplay})| \leq \phi$ 

Even if both are individually frequent under their own MIS,
the PAIR needs to satisfy the minimum MIS of the two items
AND the phi constraint simultaneously.

In our dataset:
 $\text{sup}(\text{Radiohead}) = 0.1998 \geq \text{MIS}(\text{Radiohead}) = 0.0999$  ✓
 $\text{sup}(\text{Coldplay}) = 0.1717 \geq \text{MIS}(\text{Coldplay}) = 0.0859$  ✓
 $\text{sup}(\text{pair}) = 0.0804 \geq \text{MIS}(\min) = 0.0859$  ✗ NO
 $|\text{sup difference}| = 0.0281 \leq \phi=0.05$  ✓

```

Figure 7: Figure 7 — B1 concept check output proving Radiohead and Coldplay are individually frequent but their pair is NOT frequent under MSApriori.

Itemset	Support	MIS	Frequent?
Radiohead	0.1998	0.0999	YES
Coldplay	0.1717	0.0859	YES
Radiohead + Coldplay	0.0804	0.0859	NO

Although both artists are individually frequent, their pair support (0.0804) falls below the minimum MIS required for the pair (0.0859 = MIS of Coldplay, the lower-MIS item). The phi condition is satisfied ($|0.1998 - 0.1717| = 0.0281 \leq 0.05$) but the support condition fails ($0.0804 < 0.0859$ ✗). Therefore the pair is pruned and the classmate's claim is proven

wrong with real data.

2.2 B2. Seed List L Construction [5 Marks]

To construct the seed list L, all 200 artists were sorted by MIS value in ascending order. The MSApriori seed-list algorithm scans this sorted list and adds the first item i such that $\text{sup}(i) \geq \text{MIS}(i)$, then continues adding all subsequent items also satisfying this condition. Items before the first qualifying item are excluded from L entirely — they would only generate candidates with support below the minimum threshold for the pair.

Scanning through the sorted list, the first item meeting $\text{sup} \geq \text{MIS}(\text{item})$ was **Pixies** (support = 0.0352, MIS = 0.0176). This established the entry threshold for L at MIS = 0.0176. All 200 items in the working set had support $\geq$ their own MIS (a consequence of the proportional strategy with $\alpha = 0.5$, which guarantees support = $2 \times \text{MIS}$ for every item), so all 200 were admitted to L.

```

Working with top 200 artists
Transactions with >= 2 top artists: 1666
Support range: 0.0352 to 0.3088
MIS range: 0.0176 to 0.1544

--- Seed List L ---
First item MIS threshold: 0.0176
Items in L: 200

--- F1 (Frequent 1-itemsets) ---
Items in F1: 200

Artist                Support    MIS    Pass?
-----
Pixies                0.0352    0.0176    ✓
Céline Dion           0.0358    0.0179    ✓
John Mayer            0.0358    0.0179    ✓
Girls Aloud           0.0358    0.0179    ✓
Usher                 0.0358    0.0179    ✓
The Fray              0.0358    0.0179    ✓
Jay-Z                 0.0358    0.0179    ✓
Dream Theater         0.0358    0.0179    ✓
Yann Tiersen          0.0363    0.0182    ✓
Goldfrapp             0.0363    0.0182    ✓
Bring Me The Horizon  0.0363    0.0182    ✓
...
Britney Spears        0.2642    0.1321    ✓
Lady Gaga             0.3088    0.1544    ✓

Items in L but NOT F1: 0
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```

Figure 8: Figure 8 — Seed list L construction and F1 frequent 1-itemsets. Items sorted by MIS value ascending. All 200 items in L passed their own MIS threshold.

Seed List Summary (sample):

Artist	MIS	Support	In L?
Pixies	0.0176	0.0352	Yes
Céline Dion	0.0179	0.0358	Yes
John Mayer	0.0179	0.0358	Yes
Girls Aloud	0.0179	0.0358	Yes
Usher	0.0179	0.0358	Yes
... 195 more items ...			
Britney Spears	0.1321	0.2642	Yes
Lady Gaga	0.1544	0.3088	Yes

Total items in L: **200** First item MIS threshold: **0.0176** Items in F1: **200**
 Items in L but NOT F1: **0**

All 200 items in L satisfied their own MIS threshold, resulting in $F1 = 200$ frequent 1-itemsets. This occurred because with $\alpha = 0.5$ proportional MIS, every artist's support is always exactly double their MIS value, so all items automatically pass.

Important: C2 candidates are generated from L (not F1) for generating C2 candidates. This is essential because the downward closure property is not valid under MMIS. Under standard Apriori, if a pair $\{A, B\}$ is frequent, both A and B must be in F1. Under MMIS this no longer holds — an item A can be infrequent under its own MIS yet appear in a frequent pair because the pair's MIS threshold is set by the lower-MIS partner. Using L instead of F1 prevents false pruning of valid candidates.

2.3 B3. MSApriori Execution Results [20 Marks]

Frequent 1-Itemsets (F1)

All 200 items in the seed list L form F1. Each item's support meets or exceeds its own MIS threshold (by construction under the $\alpha = 0.5$ proportional strategy). The F1 set spans Lady Gaga (support = 0.3088, MIS = 0.1544) at the popular end down to Pixies (support = 0.0352, MIS = 0.0176) at the niche boundary. No items were pruned at this level (0 pruned out of 200 candidates).

Generating C2 and F2

```

=====
PART B3: Generating C2 from L
phi = 0.05
=====

Total possible pairs:      19900
Pruned by MIS condition:    0
Pruned by phi condition:   6417
Candidates in C2:         13483
Total pruned:              6417
Pruning rate:              32.2%

--- Scanning transactions for C2 support ---

C2 candidates:    13483
Passed (F2):      176
Failed:           13307

=== F2: Frequent 2-itemsets (Top 30) ===
Itemset                                Support  MIS(min)  Pass
-----
Britney Spears + Lady Gaga              0.2389   0.1321    ✓
Rihanna + Britney Spears                 0.2203   0.1186    ✓
Katy Perry + Britney Spears              0.2053   0.1194    ✓
Christina Aguilera + Rihanna             0.1945   0.1043    ✓
Rihanna + Katy Perry                    0.1921   0.1186    ✓
Beyoncé + Rihanna                       0.1789   0.0952    ✓
Christina Aguilera + Katy Perry          0.1717   0.1043    ✓
Beyoncé + Christina Aguilera            0.1639   0.0952    ✓
Beyoncé + Katy Perry                    0.1555   0.0952    ✓
Avril Lavigne + Katy Perry               0.1507   0.0977    ✓
Madonna + Rihanna                       0.1459   0.1059    ✓
Christina Aguilera + Madonna            0.1453   0.1043    ✓
Avril Lavigne + Rihanna                  0.1399   0.0977    ✓
Beyoncé + Madonna                       0.1387   0.0952    ✓
Ke$ha + Christina Aguilera              0.1375   0.0883    ✓
Madonna + Katy Perry                    0.1309   0.1059    ✓
Avril Lavigne + Christina Aguilera       0.1279   0.0977    ✓
Shakira + Beyoncé                       0.1182   0.0773    ✓
Ke$ha + Avril Lavigne                   0.1176   0.0883    ✓
Miley Cyrus + Ke$ha                     0.1164   0.0696    ✓
Ke$ha + Beyoncé                         0.1140   0.0883    ✓
Paramore + Katy Perry                   0.1140   0.0977    ✓
Beyoncé + Avril Lavigne                 0.1122   0.0952    ✓
Kylie Minogue + Beyoncé                 0.1092   0.0729    ✓
Paramore + Avril Lavigne                0.1074   0.0977    ✓
Pink + Beyoncé                          0.1068   0.0710    ✓
Radiohead + The Beatles                 0.1068   0.0999    ✓
Shakira + Ke$ha                         0.1014   0.0773    ✓
Ke$ha + Madonna                         0.0990   0.0883    ✓
Shakira + Avril Lavigne                 0.0960   0.0773    ✓

Total F2 itemsets: 176
|

```

Figure 9: Figure 9 — C2 generation from seed list L with $\varphi = 0.05$, showing pruning statistics and top 30 frequent 2-itemsets F2.

From 19,900 total possible pairs among 200 artists:

- Pruned by MIS condition: 0
- Pruned by phi condition: 6,417

- Candidates in C2: 13,483
- Total pruned: 6,417 (32.2% pruning rate)

After scanning all 1,666 transactions, 176 pairs passed their MIS threshold forming **F2 = 176 frequent 2-itemsets**. The top pairs are dominated by mainstream pop co-listening patterns such as Britney Spears + Lady Gaga (support = 0.2389) and Rihanna + Britney Spears (support = 0.2203).

The complete F2 table (all 176 itemsets with Support, MIS(min item), and Pass status) is provided in the accompanying Jupyter notebook. The top 30 itemsets are shown in Figure 9 above; a representative sample with all required columns is given below:

Itemset	Support	MIS(min item)	Passes?
Britney Spears + Lady Gaga	0.2389	0.1321	Yes
Rihanna + Britney Spears	0.2203	0.1186	Yes
Katy Perry + Britney Spears	0.2053	0.1194	Yes
Christina Aguilera + Rihanna	0.1945	0.1043	Yes
Rihanna + Katy Perry	0.1921	0.1186	Yes
Beyoncé + Rihanna	0.1789	0.0952	Yes
Christina Aguilera + Katy Perry	0.1717	0.1043	Yes
Beyoncé + Christina Aguilera	0.1639	0.0952	Yes
Beyoncé + Katy Perry	0.1555	0.0952	Yes
Radiohead + The Beatles	0.1068	0.0999	Yes
... 166 additional itemsets in notebook ...			

Generating C3 and F3

```

=====
Generating C3 and F3
=====
F2 pairs available: 176
C3 candidates after join: 199
Pruned in prune step: 0
C3 after pruning: 199

--- Scanning transactions for C3 support ---

C3 candidates scanned: 199
F3 frequent 3-itemsets: 26

=== F3: Frequent 3-itemsets (Top 20) ===
Itemset                                     Support  MIS(min)  Pass
-----
Rihanna + Katy Perry + Britney Spears      0.1747    0.1186    ✓
Christina Aguilera + Rihanna + Katy Perry  0.1561    0.1043    ✓
Beyoncé + Christina Aguilera + Rihanna     0.1531    0.0952    ✓
Beyoncé + Rihanna + Katy Perry             0.1429    0.0952    ✓
Beyoncé + Christina Aguilera + Katy Perry  0.1327    0.0952    ✓
Christina Aguilera + Madonna + Rihanna     0.1273    0.1043    ✓
Beyoncé + Madonna + Rihanna               0.1206    0.0952    ✓
Avril Lavigne + Rihanna + Katy Perry       0.1206    0.0977    ✓
Beyoncé + Christina Aguilera + Madonna     0.1188    0.0952    ✓
Avril Lavigne + Christina Aguilera + Rihanna 0.1164    0.0977    ✓
Madonna + Rihanna + Katy Perry            0.1110    0.1059    ✓
Christina Aguilera + Madonna + Katy Perry  0.1110    0.1043    ✓
Avril Lavigne + Christina Aguilera + Katy Perry 0.1098    0.0977    ✓
Beyoncé + Madonna + Katy Perry            0.1056    0.0952    ✓
Beyoncé + Avril Lavigne + Rihanna         0.1050    0.0952    ✓
Ke$ha + Beyoncé + Christina Aguilera      0.1026    0.0883    ✓
Beyoncé + Avril Lavigne + Katy Perry      0.1002    0.0952    ✓
Beyoncé + Avril Lavigne + Christina Aguilera 0.0972    0.0952    ✓
Ke$ha + Avril Lavigne + Christina Aguilera 0.0924    0.0883    ✓
Ke$ha + Christina Aguilera + Madonna      0.0894    0.0883    ✓

=====
SUMMARY: Candidate vs Frequent Counts
=====
Level    Candidates  Frequent  Pruned
-----
F1         200         200        0
F2       13483         176     13307
F3         199          26        173

```

Figure 10: Figure 10 — F3 frequent 3-itemsets showing top 20 triplets and final summary table of candidate vs frequent counts at each level.

From 176 F2 pairs, the join step produced 199 C3 candidates. After the prune step and scanning transactions, **26 frequent 3-itemsets** were found forming F3.

Why were 0 candidates pruned in the C3 prune step? The MSApriori prune step removes any candidate 3-itemset $\{A, B, C\}$ whose 2-item subsets are not all present in F2. Since our C3 candidates were generated by joining pairs in F2 (which already passed both the MIS and phi conditions), their 2-item subsets are by construction already in F2. No additional phi or MIS violations arise at the 3-item level for these candidates, so all 199 survive the prune step. This is expected behaviour and does not indicate a bug — the pruning effect of phi is most powerful at the C2 generation stage.

The most frequent trio is **Rihanna + Katy Perry + Britney Spears** with support = 0.1747, meaning 17.47% of all 1,817 users listen to all three artists. The top F3 itemsets are

dominated by the mainstream pop cluster confirming the strong cohesion of this listener group.

Complete Summary

Level	Candidates	Frequent	Pruned
F1	200	200	0
F2	13,483	176	13,307
F3	199	26	173

The massive pruning at F2 level (13,307 pairs eliminated = 98.7% pruning rate) demonstrates the power of the MMIS framework. Without phi and MIS constraints a standard Apriori would have evaluated all 19,900 pairs and likely produced thousands of meaningless rules between incompatible artists of vastly different popularity.

3 Part C: Rules & Interpretation

3.1 C1. Association Rules (minconf = 60%) [10 Marks]

Rules were generated from all F2 and F3 frequent itemsets. For each frequent itemset, all non-empty proper subsets were tested as antecedents. A rule was kept if its confidence was at least 60%.

Before applying the confidence filter, the total number of candidate rules generated from all 402 frequent itemsets (F2 + F3) was **402 candidate rules**. After applying the 60% minimum confidence threshold, **190 rules** were retained. The remaining 212 rules were discarded as their confidence fell below the minimum actionable threshold. The full rule set was passed to Part D for scoring and selection.

The top 10 rules ranked by lift are:

```

Total frequent itemsets stored: 402
Total rules generated: 190

=== Top 10 Rules by Lift ===
# Antecedent Consequent Support Conf Lift
-----
1 Judas Priest + Iron Maiden Black Sabbath 0.0234 0.7222 12.7406
2 Judas Priest Black Sabbath 0.0282 0.7119 12.5592
3 Slayer Megadeth 0.0228 0.6186 12.4882
4 Selena Gomez & the Scene Demi Lovato 0.0426 0.6296 10.3054
5 Demi Lovato Selena Gomez & the Scene 0.0426 0.6976 10.3054
6 Judas Priest + AC/DC Iron Maiden 0.0216 0.8780 10.0976
7 Judas Priest + Black Sabbath Iron Maiden 0.0234 0.8298 9.5426
8 Judas Priest Iron Maiden 0.0324 0.8180 9.4067
9 Jordin Sparks Leona Lewis 0.0318 0.7317 9.2971
10 Megadeth Iron Maiden 0.0396 0.7998 9.1977

Full rule set saved for Part D

```

Figure 11: Figure 11 — Top 10 association rules by lift with support, confidence and lift values.

#	Antecedent	Consequent	Sup	Conf	Lift
1	Judas Priest + Iron Maiden	Black Sabbath	0.0234	0.7222	12.74
2	Judas Priest	Black Sabbath	0.0282	0.7119	12.56
3	Slayer	Megadeth	0.0228	0.6186	12.49
4	Selena Gomez & the Scene	Demi Lovato	0.0426	0.6296	10.31
5	Demi Lovato	Selena Gomez & the Scene	0.0426	0.6976	10.31
6	Judas Priest + AC/DC	Iron Maiden	0.0216	0.8780	10.10
7	Judas Priest + Black Sabbath	Iron Maiden	0.0234	0.8298	9.54
8	Judas Priest	Iron Maiden	0.0324	0.8180	9.41
9	Jordin Sparks	Leona Lewis	0.0318	0.7317	9.30
10	Megadeth	Iron Maiden	0.0396	0.7998	9.20

All top 10 rules have lift values significantly above 1.0, ranging from 9.20 to 12.74, confirming these are extremely strong non-random associations. The metal cluster (Judas Priest, Iron Maiden, Black Sabbath, Megadeth, Slayer) produces the strongest rules in the entire dataset. The Selena Gomez and Demi Lovato pairing (lift = 10.31) reflects a shared Disney Channel fanbase where both artists built their careers simultaneously. The Jordin Sparks and Leona Lewis pairing (lift = 9.30) reflects listeners who follow talent competition show winners as a cohesive group.

3.2 C2. Listener Personas [12 Marks]

PART C2: Listener Personas					
Top 30 Rules by Lift (for persona analysis):					
#	Antecedent	Consequent	Support	Conf	Lift
1	Judas Priest + Iron Maiden	Black Sabbath	0.0234	0.7222	12.7400
2	Judas Priest	Black Sabbath	0.0282	0.7119	12.5592
3	Slayer	Megadeth	0.0228	0.6180	12.4882
4	Selena Gomez & the Scene	Deni Lovato	0.0420	0.6290	10.3054
5	Deni Lovato	Selena Gomez & the Scene	0.0420	0.6970	10.3054
6	Judas Priest + AC/DC	Iron Maiden	0.0210	0.8780	10.0970
7	Judas Priest + Black Sabbath	Iron Maiden	0.0234	0.8298	9.5420
8	Judas Priest	Iron Maiden	0.0324	0.8180	9.4007
9	Jordin Sparks	Leona Lewis	0.0318	0.7317	9.2971
10	Megadeth	Iron Maiden	0.0390	0.7998	9.1977
11	KISS	Iron Maiden	0.0282	0.7429	8.5433
12	Natasha Bedingfield	Leona Lewis	0.0258	0.6700	8.5127
13	Nicole Scherzinger	Jennifer Lopez	0.0300	0.7790	8.3264
14	Black Sabbath	Iron Maiden	0.0402	0.7094	8.1580
15	Judas Priest + Iron Maiden	AC/DC	0.0210	0.6607	8.0221
16	The Who	Led Zeppelin	0.0300	0.7131	7.9492
17	Sugababes	Leona Lewis	0.0240	0.6144	7.8073
18	Hilary Duff	Ashley Tisdale	0.0504	0.6591	7.6278
19	Deni Lovato	Ashley Tisdale	0.0402	0.6583	7.6188
20	Judas Priest	AC/DC	0.0240	0.6211	7.4732
21	Ciara	Jennifer Lopez	0.0312	0.6752	7.2162
22	Pixie Lott	Ashley Tisdale	0.0318	0.6085	7.0419
23	Los Hermanos	The Strokes	0.0300	0.6197	7.0372
24	Jordin Sparks	Ashley Tisdale	0.0204	0.6074	7.0301
25	Janet Jackson	Jennifer Lopez	0.0288	0.6544	6.9942
26	Selena Gomez & the Scene	Ashley Tisdale	0.0408	0.6030	6.9781
27	The Doors	Led Zeppelin	0.0300	0.6100	6.8668
28	Black Sabbath	Led Zeppelin	0.0348	0.6141	6.8460
29	Jennifer Lopez	The Pussycat Dolls	0.0612	0.6544	6.1928
30	Leona Lewis	Mariah Carey	0.0540	0.6940	6.1510
--- Rules grouped by pattern ---					
🔥 METAL/ROCK Rules:					
Judas Priest + Iron Maiden - Black Sabbath (conf=0.72, lift=12.74)					
Judas Priest - Black Sabbath (conf=0.71, lift=12.56)					
Slayer - Megadeth (conf=0.62, lift=12.49)					
Judas Priest + AC/DC - Iron Maiden (conf=0.88, lift=10.10)					
Judas Priest + Black Sabbath - Iron Maiden (conf=0.83, lift=9.54)					
Judas Priest - Iron Maiden (conf=0.82, lift=9.41)					
Megadeth - Iron Maiden (conf=0.80, lift=9.20)					
KISS - Iron Maiden (conf=0.74, lift=8.54)					
🎤 POP QUEENS Rules:					
Pink + Beyoncé - Shakira (conf=0.72, lift=4.69)					
Ke\$ha + Beyoncé - Shakira (conf=0.72, lift=4.66)					
Shakira + Beyoncé - Pink (conf=0.65, lift=4.61)					
Ke\$ha + Beyoncé - Pink (conf=0.63, lift=4.45)					
Beyoncé - Madonna + Rihanna (conf=0.63, lift=4.34)					
Madonna + Rihanna - Beyoncé (conf=0.83, lift=4.34)					
Pink + Shakira - Beyoncé (conf=0.83, lift=4.34)					
Ke\$ha + Madonna - Christina Aguilera (conf=0.90, lift=4.33)					
★ TEEN/INDIE-POP Rules:					
Selena Gomez & the Scene - Deni Lovato (conf=0.63, lift=10.31)					
Deni Lovato - Selena Gomez & the Scene (conf=0.70, lift=10.31)					
Deni Lovato - Ashley Tisdale (conf=0.66, lift=7.62)					
Selena Gomez & the Scene - Ashley Tisdale (conf=0.60, lift=6.98)					
Miley Cyrus - Taylor Swift (conf=0.60, lift=5.03)					
Taylor Swift - Miley Cyrus (conf=0.70, lift=5.03)					
Ke\$ha - Miley Cyrus (conf=0.60, lift=4.73)					
Miley Cyrus - Ke\$ha (conf=0.84, lift=4.73)					
--- Support Stats for Persona Artists ---					
Artist	Support	Users			
			18		
Metallica	0.1002	193			
Iron Maiden	0.0870	158			
Black Sabbath	0.0507	103			
Judas Priest	0.0390	72			

Three distinct listener personas were identified from the association rules:

Persona 1: “The Metal Pilgrim”

Profile: This listener is deeply rooted in classic and thrash metal. Their library is built around the founding bands of heavy metal — artists who defined the genre across the 1970s to 1990s. They do not listen to metal casually — they collect it systematically.

Defining Rules (both derived from F3-level frequent itemsets):

- Rule 1: Judas Priest + Iron Maiden $\rightarrow$ Black Sabbath (conf = 72.2%, lift = 12.74)
Source itemset: {Judas Priest, Iron Maiden, Black Sabbath} $\in$ F3, support = 0.0234
- Rule 2: Slayer $\rightarrow$ Megadeth (conf = 61.9%, lift = 12.49)
Source itemset: {Slayer, Megadeth} $\in$ F2, support = 0.0228

Interpretation: A user listening to both Judas Priest and Iron Maiden has a 72% chance of also listening to Black Sabbath — 12.74 times more likely than random chance. This is the single strongest rule in our entire dataset. The Slayer $\rightarrow$ Megadeth rule (lift = 12.49) confirms the thrash metal sub-community is equally tight. These listeners show almost tribal loyalty to their genre with confidence values consistently above 70%.

Recommendation: A “**Metal Foundations**” playlist featuring Black Sabbath, Ozzy Osbourne, Deep Purple and Dio — completing the classic metal canon for this persona and introducing related artists they are statistically very likely to enjoy.

Persona 2: “The Pop Queen Devotee”

Profile: This listener follows the dominant female pop artists of the 2000s–2010s as a cohesive group rather than individually. Their library contains multiple pop queens simultaneously, suggesting they follow the era rather than any single artist.

Defining Rules (both derived from F3-level frequent itemsets):

- Rule 1: Madonna + Rihanna $\rightarrow$ Beyoncé (conf = 83%, lift = 4.34)
Source itemset: {Madonna, Rihanna, Beyoncé} $\in$ F3, support = 0.1206
- Rule 2: Ke\$ha + Madonna $\rightarrow$ Christina Aguilera (conf = 90%, lift = 4.33)
Source itemset: {Ke\$ha, Madonna, Christina Aguilera} $\in$ F3, support = 0.0894

Interpretation: Users who listen to both Madonna and Rihanna have an 83% probability of also listening to Beyoncé. The 90% confidence on Ke\$ha + Madonna $\rightarrow$ Christina Aguilera is remarkably high — nearly every single user with both Ke\$ha and Madonna also has Christina Aguilera in their library. These artists function as a cluster rather than individual choices for this persona.

Recommendation: A “**Queens of Pop**” Discovery Mix seeded with deep cuts from Beyoncé, Christina Aguilera and Madonna — moving beyond their biggest hits to explore album tracks and collaborations that this persona has not yet discovered.

Persona 3: “The Disney Generation Fan”

Profile: This listener grew up with Disney Channel artists and maintains strong loyalty to that entire cohort. Their listening patterns reflect shared cultural nostalgia more than pure musical similarity — they follow the people, not just the sound.

Defining Rules (both derived from F2-level frequent itemsets):

- Rule 1: Selena Gomez & the Scene $\rightarrow$ Demi Lovato (conf = 63%, lift = 10.31)
Source itemset: {Selena Gomez & the Scene, Demi Lovato} $\in$ F2, support = 0.0426
- Rule 2: Demi Lovato $\rightarrow$ Ashley Tisdale (conf = 65.8%, lift = 7.62)
Source itemset: {Demi Lovato, Ashley Tisdale} $\in$ F2, support = 0.0402

Interpretation: Selena Gomez and Demi Lovato show a lift of 10.31 — one of the highest values in our entire dataset — meaning their co-occurrence is over 10 times more likely than random chance. This reflects a fanbase that followed both artists together from their Disney origins through their solo careers. The chain extends further to Ashley Tisdale (lift = 7.62) confirming a broader Disney ecosystem of listening that spans the entire channel’s artist roster.

Recommendation: A “**Disney Era Throwback**” playlist featuring Miley Cyrus, Hilary Duff, Ashley Tisdale and Jonas Brothers — completing the nostalgia collection for this persona and surfacing artists from the same era they are statistically very likely to enjoy.

4 Part D: Design Challenge

4.1 D1. Rule Scoring Function [12 Marks]

To select the best 50 rules for the Discovery Mix engine from 190 total rules, a composite scoring function was designed combining three metrics:

$$\text{Score} = 0.2 \times \text{support} + 0.3 \times \text{confidence} + 0.5 \times \text{normalized_lift}$$

Weight justification:

- **Support (weight = 0.2):** Given the lowest weight because many strong niche rules have naturally low support — penalizing them too heavily would bias the entire engine toward mainstream artists only and defeat the purpose of using MMIS
- **Confidence (weight = 0.3):** Given medium weight because it ensures the rule fires reliably when triggered by a matching user — a rule with very low confidence would produce poor recommendations even if its lift is high
- **Lift (weight = 0.5):** Given the highest weight because lift measures how surprising and non-random a rule is, which is the most valuable property for a discovery recommendation engine — we want to recommend artists the user would not have found on their own

Lift was normalized to the 0–1 range before scoring using min-max normalization to ensure fair combination with support and confidence which are already naturally in the 0–1 range.

```

1 =====
2 PART D1: Rule Scoring Function
3 Score = 0.2*support + 0.3*confidence + 0.5*norm_lift
4 =====
5
6 Total rules before deduplication: 190
7 After removing redundant rules: 76
8
9 === Top 15 Rules by Score ===
10 # Antecedent Consequent Sup Conf Lift Score
11 -----
12 1 Judas Priest + Iron Maiden Black Sabbath 0.0234 0.7222 12.7406 0.7213
13 2 Judas Priest Black Sabbath 0.0282 0.7119 12.5592 0.7103
14 3 Slayer Megadeth 0.0228 0.6186 12.4882 0.6777
15 4 Judas Priest + AC/DC Iron Maiden 0.0216 0.8780 10.0976 0.6376
16 5 Demi Lovato Selena Gomez & the Scene 0.0426 0.6976 10.3054 0.5979
17 6 Judas Priest + Black Sabbath Iron Maiden 0.0234 0.8298 9.5426 0.5962
18 8 Selena Gomez & the Scene Demi Lovato 0.0426 0.6296 10.3054 0.5775
19 9 Megadeth Iron Maiden 0.0396 0.7998 9.1977 0.5734
20 10 Jordin Sparks Leona Lewis 0.0318 0.7317 9.2971 0.5563
21 11 Nicole Scherzinger Jennifer Lopez 0.0360 0.7790 8.3264 0.5236
22 12 KISS Iron Maiden 0.0282 0.7429 8.5433 0.5219
23 13 Natasha Bedingfield Leona Lewis 0.0258 0.6700 8.5127 0.4980
24 14 Black Sabbath Iron Maiden 0.0402 0.7094 8.1586 0.4953
25 15 The Who Led Zeppelin 0.0306 0.7131 7.9492 0.4841
26 17 Hilary Duff Ashley Tisdale 0.0504 0.6591 7.6278 0.4561
27
28 =====
29 PART D2: Cold Start Recommendation Demo
30 =====
31
32 New user listens to: Judas Priest + Iron Maiden
33 Top 5 recommendations:
34 Recommended Because Conf Lift Score
35 -----
36 Black Sabbath Judas Priest + Iron Maiden 0.7222 12.7406 0.7213
37 AC/DC Judas Priest + Iron Maiden 0.6667 8.0221 0.4720
38
39 New user listens to: Selena Gomez & the Scene + Demi Lovato
40 Top 5 recommendations:
41 Recommended Because Conf Lift Score
42 -----
43 Ashley Tisdale Demi Lovato 0.6583 7.6188 0.4534
44
45 New user listens to: Rihanna + Beyoncé
46 Top 5 recommendations:
47 Recommended Because Conf Lift Score
48 -----
49 Britney Spears Rihanna 0.9287 3.5155 0.3685
50 Christina Aguilera Beyoncé 0.8605 4.1256 0.3668
51 Katy Perry Rihanna 0.8098 3.3901 0.3210
52 Madonna Beyoncé 0.6336 4.3438 0.3008
53 Shakira Beyoncé 0.6210 4.0153 0.2803
54
55 =====
56 FINAL: Top 50 Rules for Discovery Mix Engine
57 =====
58 Selected 50 rules for inference
59
60 Score range: 0.3423 to 0.7213
61 Lift range: 3.4424 to 12.7406
62 Conf range: 0.6065 to 0.9393
63 |

```

Figure 13: Figure 13 — Top 15 scored rules after applying scoring function, deduplication results, and cold start recommendation examples for three different new user scenarios.

Top 15 Scored Rules:

#	Antecedent	Consequent	Sup	Conf	Lift	Score
1	Judas Priest + Iron Maiden	Black Sabbath	0.0234	0.7222	12.74	0.7213
2	Judas Priest	Black Sabbath	0.0282	0.7119	12.56	0.7103
3	Slayer	Megadeth	0.0228	0.6186	12.49	0.6777
4	Judas Priest + AC/DC	Iron Maiden	0.0216	0.8780	10.10	0.6376
5	Demi Lovato	Selena Gomez & the Scene	0.0426	0.6976	10.31	0.5979
6	Judas Priest + Black Sabbath	Iron Maiden	0.0234	0.8298	9.54	0.5962
7	Selena Gomez & the Scene	Demi Lovato	0.0426	0.6296	10.31	0.5775
8	Megadeth	Iron Maiden	0.0396	0.7998	9.20	0.5734
9	Jordin Sparks	Leona Lewis	0.0318	0.7317	9.30	0.5563
10	Nicole Scherzinger	Jennifer Lopez	0.0360	0.7790	8.33	0.5236
11	KISS	Iron Maiden	0.0282	0.7429	8.54	0.5219
12	Natasha Bedingfield	Leona Lewis	0.0258	0.6700	8.51	0.4980
13	Black Sabbath	Iron Maiden	0.0402	0.7094	8.16	0.4953
14	The Who	Led Zeppelin	0.0306	0.7131	7.95	0.4841
15	Hilary Duff	Ashley Tisdale	0.0504	0.6591	7.63	0.4561

Redundancy Removal: A rule R_1 was defined as redundant with respect to a higher-scoring rule R_2 if both share the same antecedent and the same consequent direction (i.e., R_1 is a less specific or lower-scoring version of R_2). Specifically, for each unique antecedent, only the single highest-scoring rule was retained. This definition was chosen because in an inference engine with a 50-rule budget, serving two rules with the same trigger antecedent wastes a slot that could cover a different user context. After applying this deduplication, 190 rules reduced to **76 unique rules**. The top 50 were selected for the Discovery Mix engine:

- Score range: 0.3423 to 0.7213
- Lift range: 3.4424 to 12.7406
- Confidence range: 60.65% to 93.93%

The top scoring rule (Judas Priest + Iron Maiden $\rightarrow$ Black Sabbath, score = 0.7213) ranks first because it combines high confidence (72.2%) with the highest lift in the dataset (12.74), making it both reliable and highly surprising — exactly what a discovery engine needs.

4.2 D2. Cold Start Recommendation Strategy [8 Marks]

When a new user signs up and listens to exactly 2 artists, the recommendation engine operates using the following algorithm:

1. Search all 50 rules where the antecedent is a subset of the user's 2 artists
2. Collect all consequent artists from every matching rule
3. Remove any artists the user already listens to from the recommendation list
4. Rank remaining recommendations by rule score in descending order
5. Return the top N unique artist recommendations

This approach is lightweight (only 50 rules evaluated at inference time), fully interpretable, and immediately actionable from just 2 data points about a new user.

Example 1 — New user listens to: Judas Priest + Iron Maiden

Recommended	Triggering Rule	Confidence	Lift	Score
Black Sabbath	Judas Priest + Iron Maiden	72.2%	12.74	0.721
AC/DC	Judas Priest + Iron Maiden	66.7%	8.02	0.472

Example 2 — New user listens to: Selena Gomez & the Scene + Demi Lovato

Recommended	Triggering Rule	Confidence	Lift	Score
Ashley Tisdale	Demi Lovato	65.8%	7.62	0.453

Example 3 — New user listens to: Rihanna + Beyoncé

Recommended	Triggering Rule	Confidence	Lift	Score
Britney Spears	Rihanna	92.9%	3.52	0.369
Christina Aguilera	Beyoncé	86.1%	4.13	0.367
Katy Perry	Rihanna	81.0%	3.39	0.321
Madonna	Beyoncé	63.4%	4.34	0.301
Shakira	Beyoncé	62.1%	4.02	0.280

The three examples demonstrate the system working correctly across all three listener personas identified in Part C — metal fans get metal recommendations, Disney fans get Disney recommendations, and pop fans get pop recommendations. The engine is persona-aware without ever explicitly labelling the user.